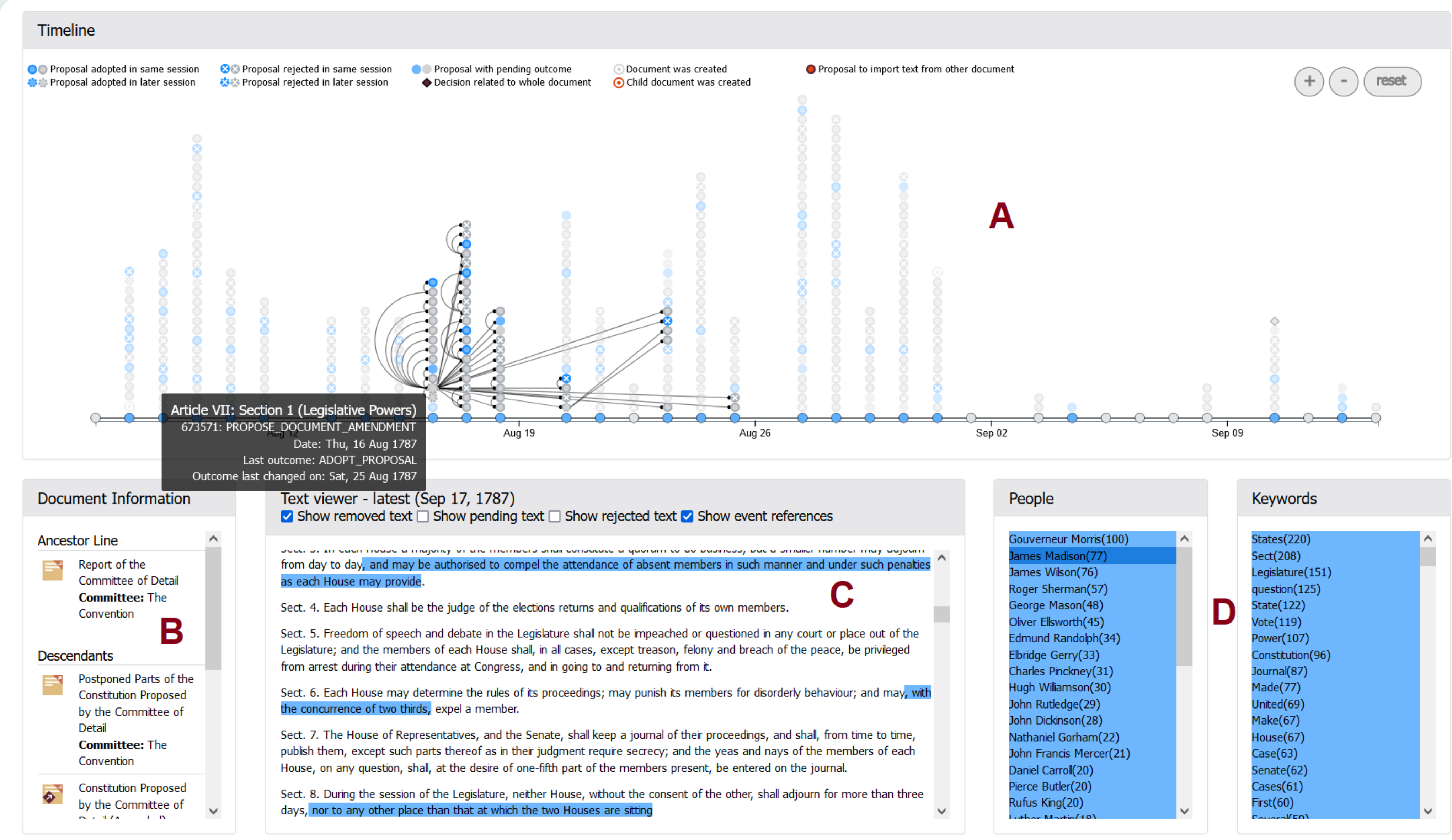


Hierarchical Timeline Exploration for Collaborative Text Writing

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Interface for exploring text using the timeline. The timeline itself (A) shows a history of proposals for amendments to the text, which may have been approved or not. These proposals can be further examined to display event details and relationships to other proposals. A list of related documents (B) shows other documents that had text copied to or from the current document. The text itself is shown in a central position (C), which can be set to include removed or rejected text. Finally, lists of people and keywords (D) involved in the document are shown. Clicking any part of the text in C or any element in D will make all components of the visualization highlight content related to what was clicked.

Introduction

Collaborative text development can take many forms, from source code management to building and maintaining knowledge repositories. Several actors propose and enact changes with different protocols for approval, and change histories are often structured as hierarchical sequences of events, where each proposed alteration can itself receive further alteration proposals. Additionally, each application domain contains specific operations and relationships between authors, metadata objects, and the text itself.

We present an approach to tackle this challenge, focusing on historical records of foundational legal texts in humanities, designed and developed alongside domain researchers that work closely with this data. This poster focuses on exploring the timeline, which is a component of a larger analytical tool under development.

The Quill Project

The Quill Project [1] is a research initiative that investigates text analysis and data visualization containing well-organized, event-based datasets representing the negotiation of historical legal texts. These datasets are built by Quill researchers from scanned versions of texts, minutes, and other records of a negotiation from historical archives, which are converted into text and used to build event sequences that are stored in relational databases.

These sequences can quickly grow in size and complexity, which prompts our research into visualization methods to aid Quill researchers in navigating and finding relevant information in such datasets.

Design

Quill document timeline sequences pose a particular set of challenges: namely, that each negotiation has its own set of procedures to enact a change to a text, and that metadata for each event may contain several interconnected elements – dates, people, keywords, and so on.

When discussing a possible visualization solution and its requirements with Quill researchers, one task was often mentioned: visualizing **provenance** for sections of a text. That is, for a given sentence or paragraph of a document, researchers wanted to see when it was discussed or altered, who was involved in the discussions, what topics are relevant to it, and an overall timeline of events that led to its current state.

To achieve this goal, we defined 3 challenges our approach needed to tackle: how to summarize the contents of the event timeline, how to handle multiple hierarchies at the same time, and how to display relationships between events and multiple entities in the dataset.

Our approach was to build an interactive timeline containing amendment proposals over time, connected to the text itself so that selections made in one of them are reflected in the other.

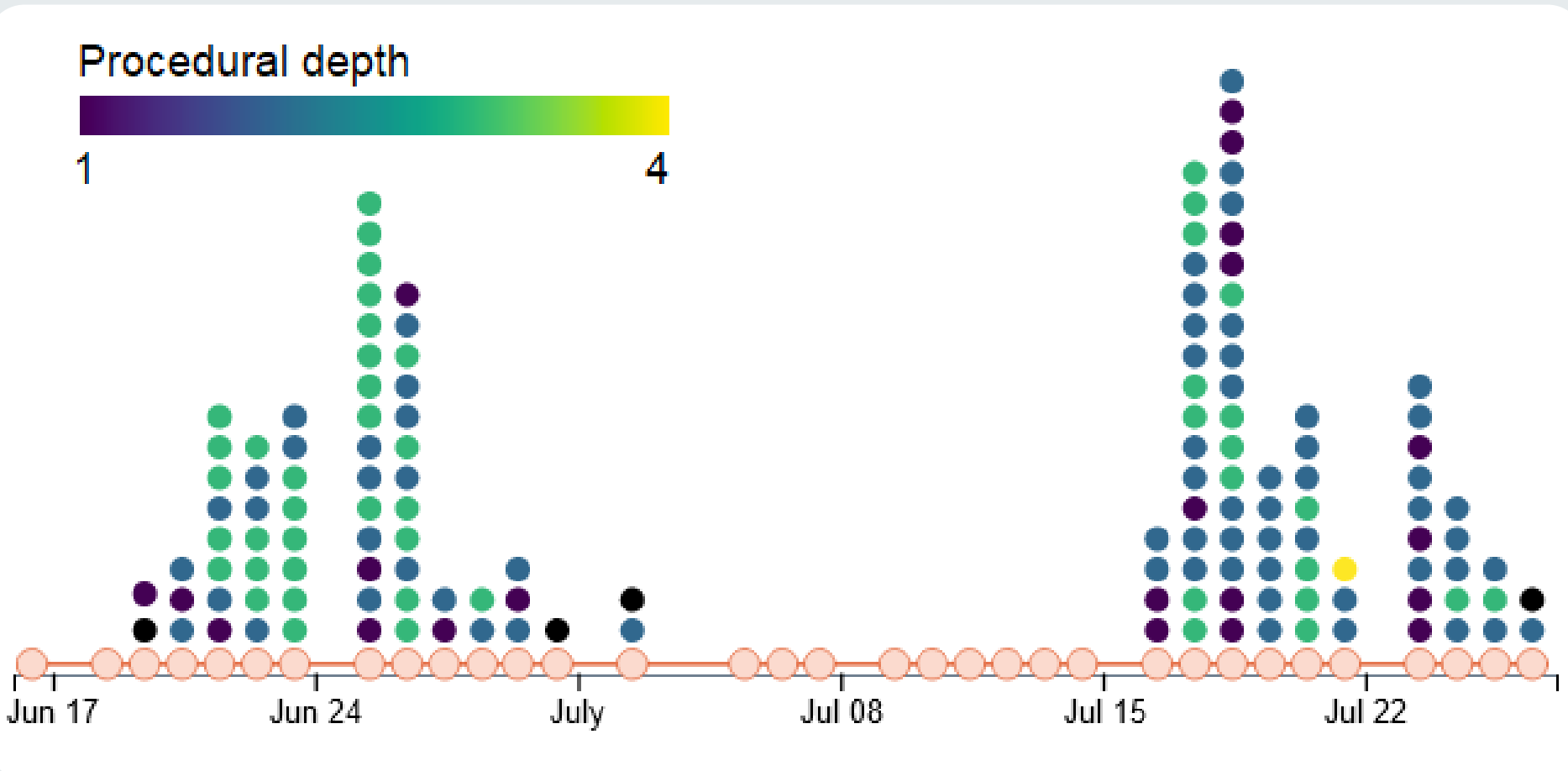
Summarization

We needed a compact representation for events that was both scalable and of easy interpretation as to provide a notion of the activity in the timeline at a quick glance. To achieve this, we developed a symbol system that concatenates the entire event sequence of proposing an amendment, discussing it and deciding it into a single symbol. This representation needed to be intuitive for researchers already working with other interfaces in Quill, so we employed a similar color scheme.

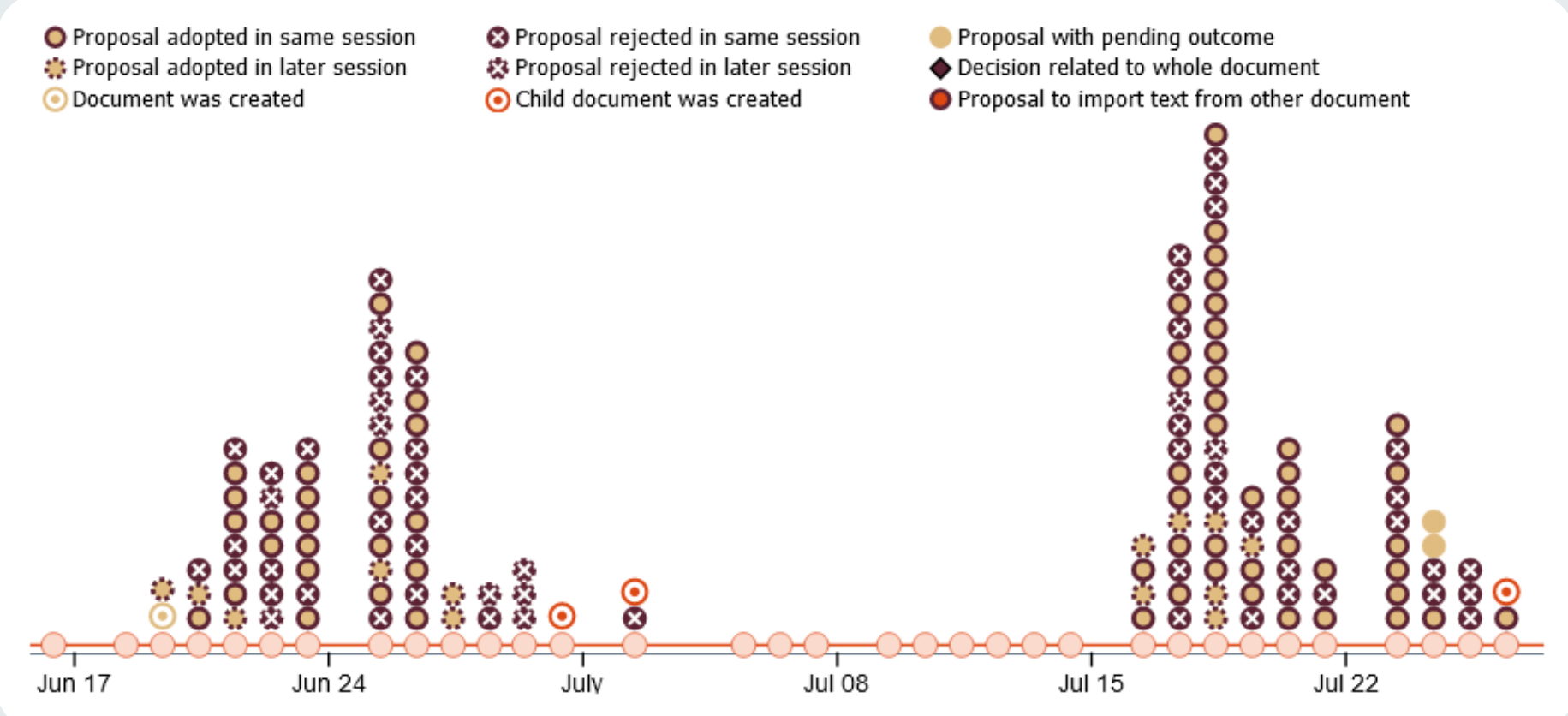


Multiple Hierarchies

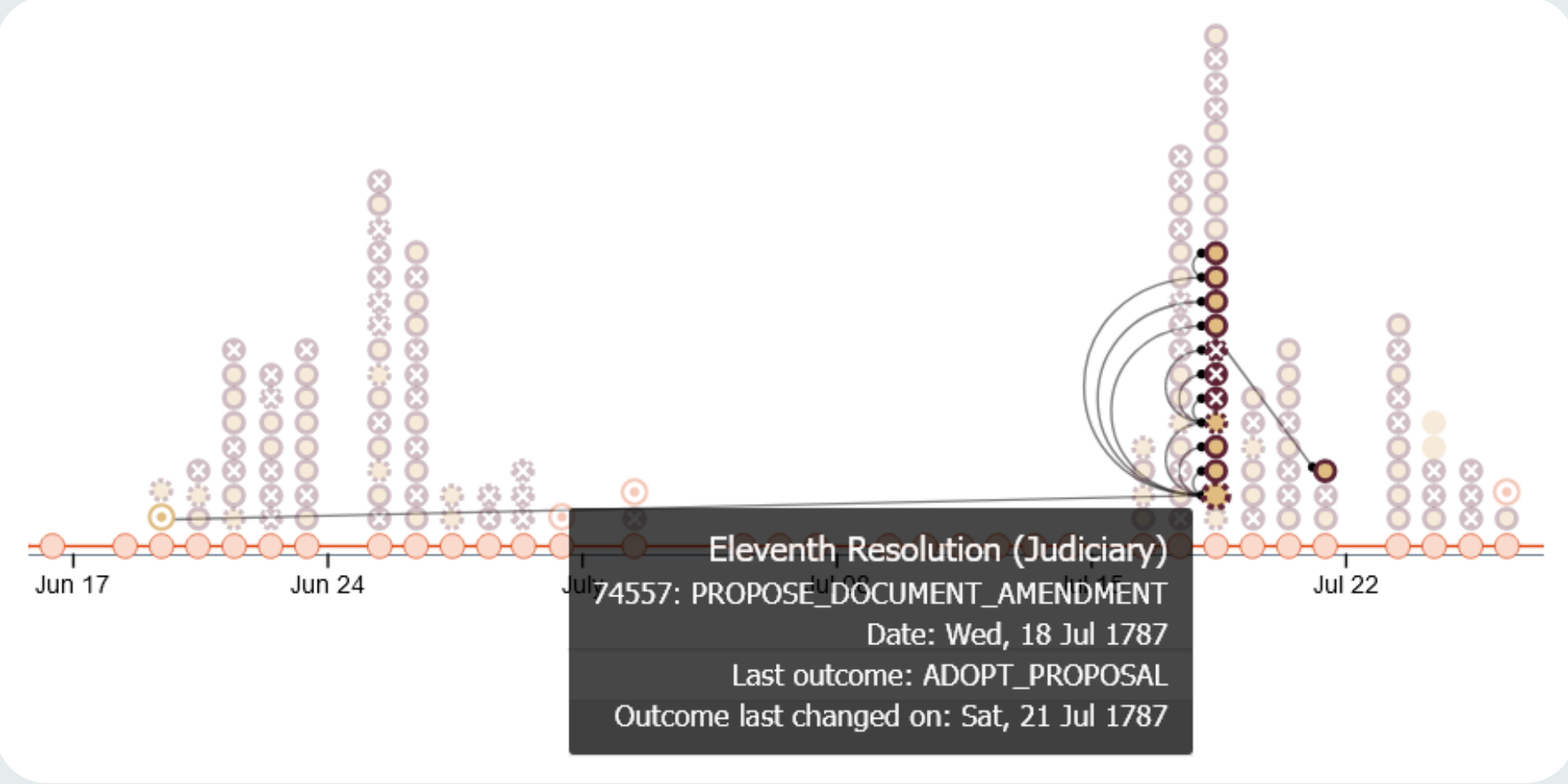
Our timeline representation needed to display multiple hierarchical structures at the same time: temporal (events are contained within sessions, which are contained within committees), procedural (an event may occur as a development of another event, and so on). We use geometric positioning to portray the first, and on-demand arcs connecting symbols to display procedural relationships.



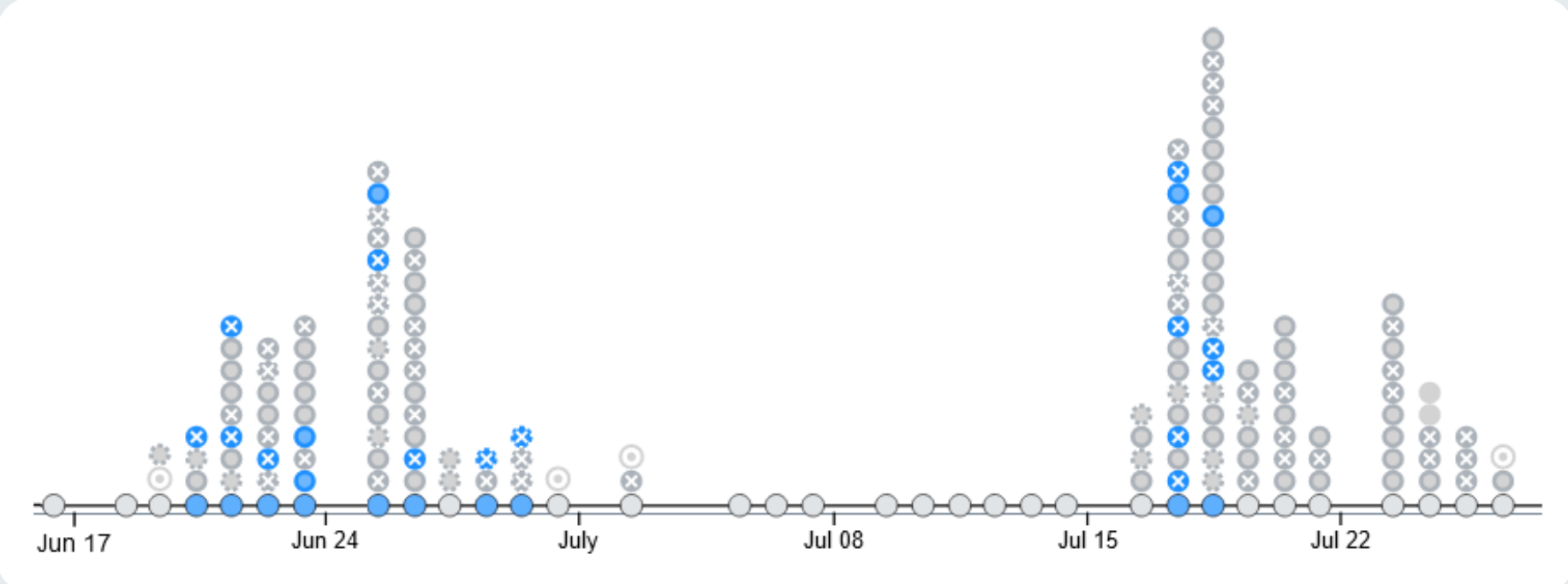
This image shows the different levels of procedural depth for each proposal in a timeline.



Default representation for timeline



Hovering over icon shows details and procedural descendants



Selection mode

Entity Relationships

To visualize relationships between people, topics, proposals, and text, we employ a selection system: users can select either a text fragment, a person’s name, a keyword, or a proposal: the visualization will enter “selection mode”, and every component will change as to highlight content related to the selected element in blue. For instance, selecting a part of the text will highlight all people involved in proposing and deciding it, all related keywords, all events in the timeline that contributed to it, and other parts of the text that were decided in the same proposals.

Preliminary Results

Preliminary feedback from domain experts indicates that our toolset was effective in communicating the contents of a document timeline and in displaying provenance for text fragments. Further evaluation, including a formal study with domain experts, will be conducted as future work.

Conclusions

This poster presented a visualization approach to display event history timelines for collaborative text development, applied to historical legal documents. This approach is developed alongside humanities experts, who provide domain requirements and feedback along the process. Our approach still presents challenges for future work, such as tracking the evolution of a document as it is copied and transferred along multiple timelines, and further improving summarization and scalability, by grouping proposals over time and providing visual summaries of text changes and interactions.

References

[1] N. Cole, A. Abdul-Rahman, and G. Mallon. Quill: A framework for constructing negotiated texts – with a case study on the US Constitutional Convention of 1787. In 2017 ACM/IEEE Joint Conference on Digital Libraries (JCDL), pp. 1–10. IEEE, 2017. doi: 10.1109/JCDL.2017.7991562

